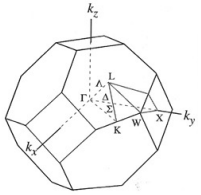
# Crystal Lattices

Baile Chen

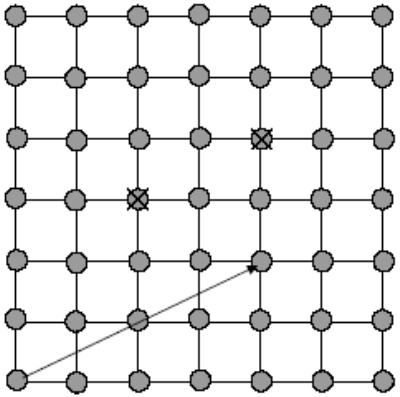
School of Information Science and Technology

Shanghai Tech University



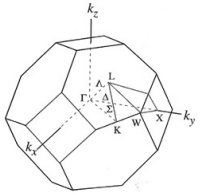
# Semiconductors

- Group IV elements: Si, Ge, C
- Compound Semiconductors :
  - III-V (GaAs, InP, AlAs)
  - II-VI (ZnSe, CdS)
- Tertiary (InGaAs,AlGaAs)
- Quaternary (InGaAsP)

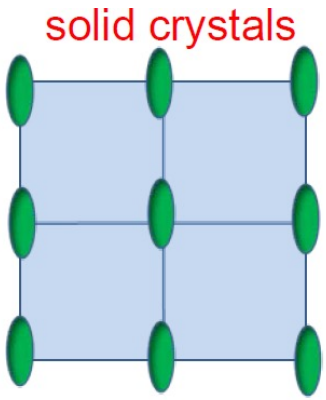


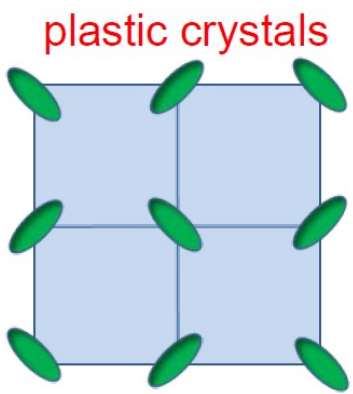
| III | IV | V |
|---|---|---|
| B | C | |
| Al | Si | P |
| Ga | Ge | As |
| In | | Sb |

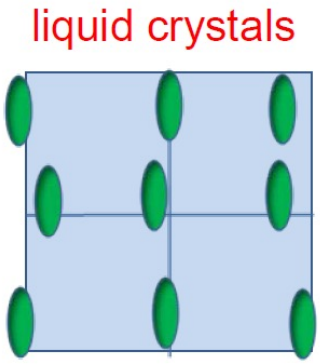
**Not all combinations possible:**
**Lattice mismatch, room temp. instability, etc are concerns**

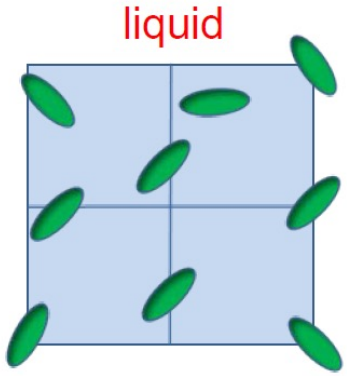


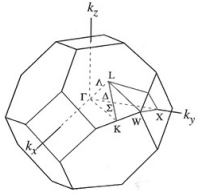
# Arrangement of Atoms: orientation vs. position



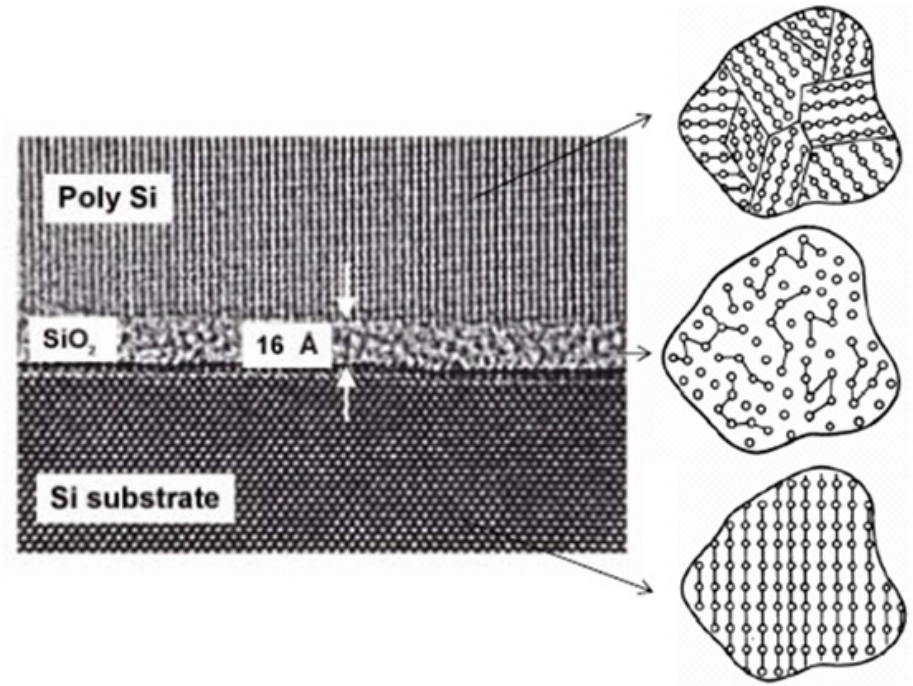



# Solids

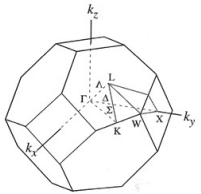



Poly-crystalline
Thin Film
Transistors

Amorphous
Oxides.

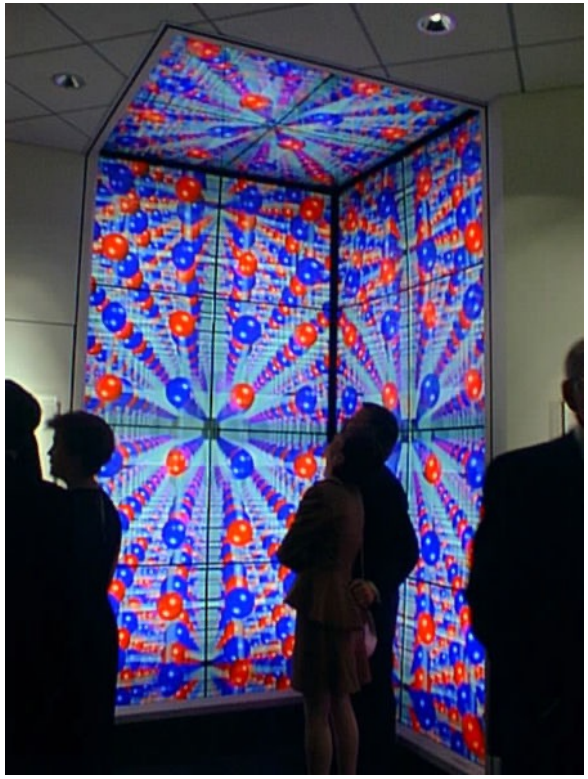
Crystalline.
*Definition ?*

**Metals**: Gates, Interconnects



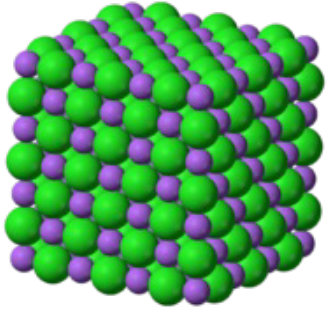
# Solids Tend to Form Ordered Crystals

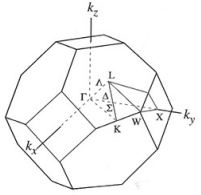
Natural History Museum, DC



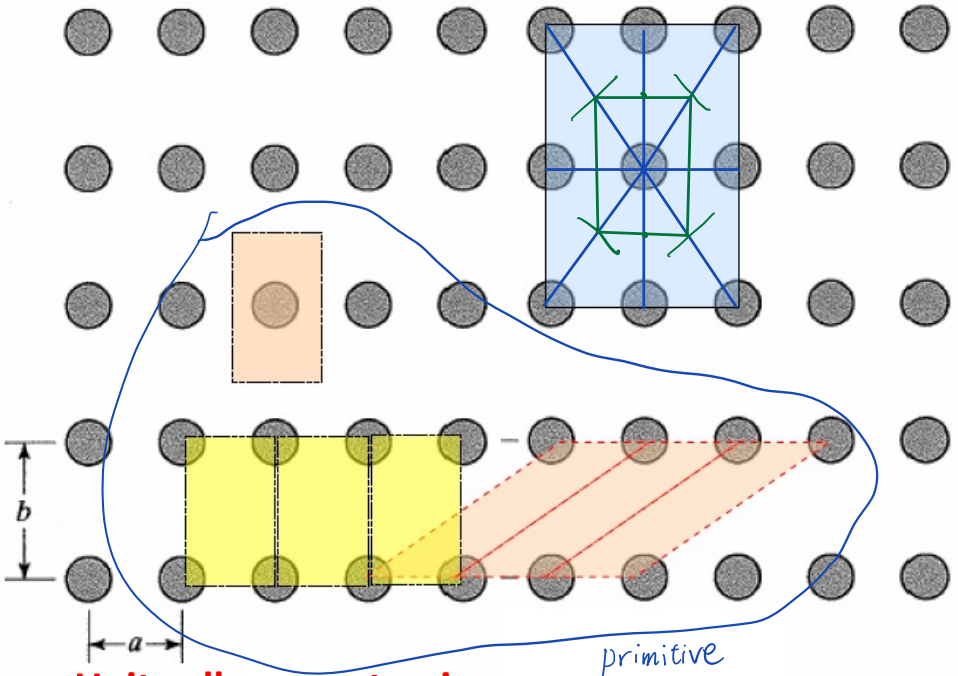


(Rock salt, NaCl)

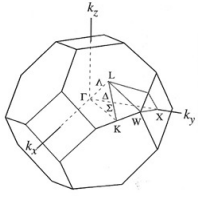




# Unit cell of Periodic Lattice

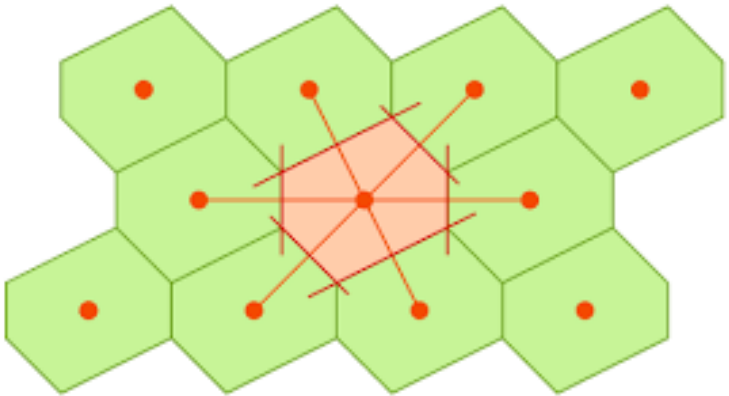



- **Unit cells are not unique**
- **Unit cells can be primitive and Non-primitive**

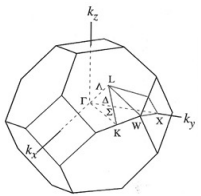


# Wigner-Seitz Primitive Cell

- Choose a reference atom
- Connect to all its neighbors by straight lines
- Draw lines (in 2D) or planes (in 3D) normal to and at the midpoints of lines drawn in step 2
- Smallest volumes enclosed is the Wigner-Seitz primitive cell



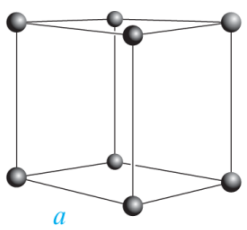
http://www.physics-in-a-nutshell.com/article/5



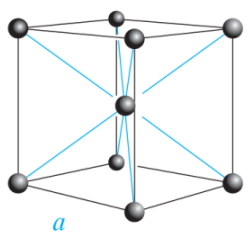
# Stuffing Atoms into Unit Cells

- How many atoms per unit cell?
- Avogadro's number: $N_A$ = # atoms / mole
- Atomic mass: A = grams / mole
- Atom counting in unit cell: atoms / $cm^3$
- How do you calculate density?

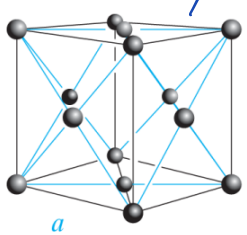
$$\rho = \frac{4 \cdot \frac{A}{N_A}}{a^3}$$



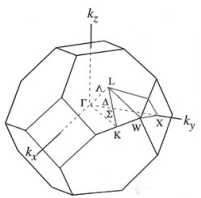
Simple cubic



Body-centered cubic



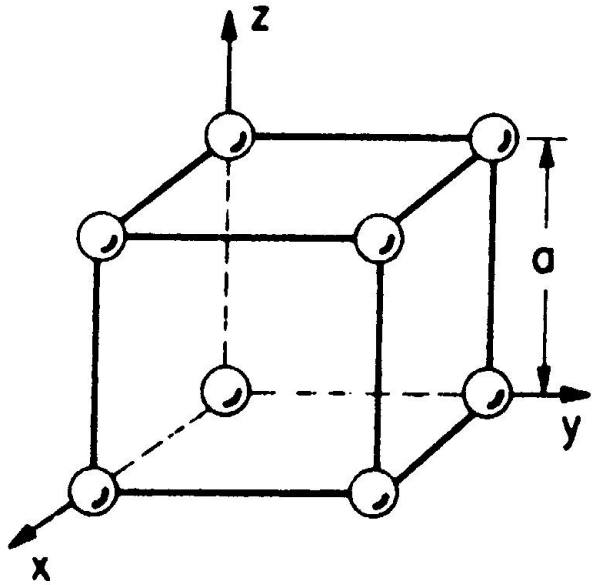
Face-centered cubic

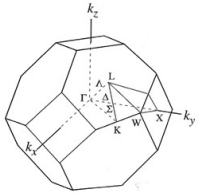


# Simple Cubic Structure

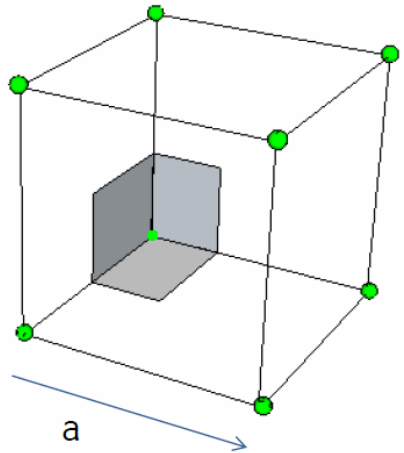
- Coordination Number (# of nearest nbs. = ?) 6
- # of atoms/cell = ? 1
- Packing fraction = ?

$$\frac{\frac{4}{3}\pi r^3}{a^3} \quad 2r = a$$



# Cubic Lattice: Volume Issues



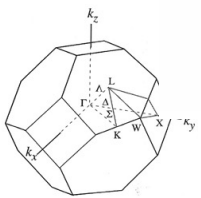

**Points per cell**

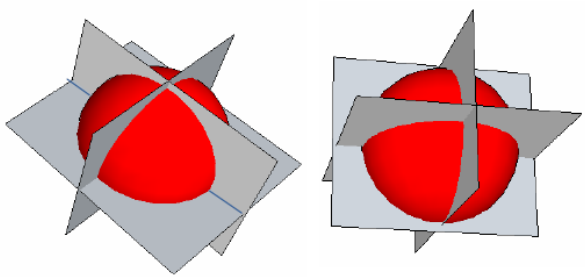
=1/8 points/corner x 8 corners

=1 Point/cell

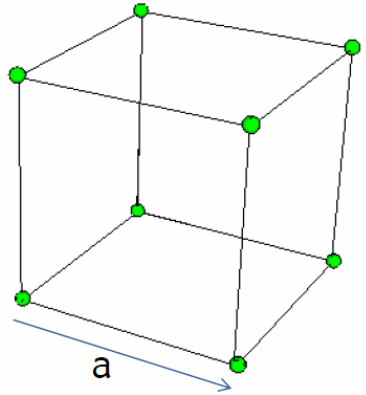
**Number density**

$=(1/a^3)$ points/$cm^3$





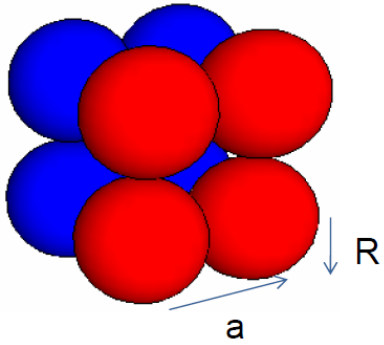
# Cubic Lattice: Volume Issues

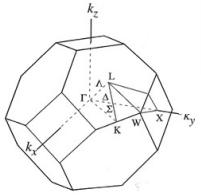



**Packing density=volume filled/total volume**

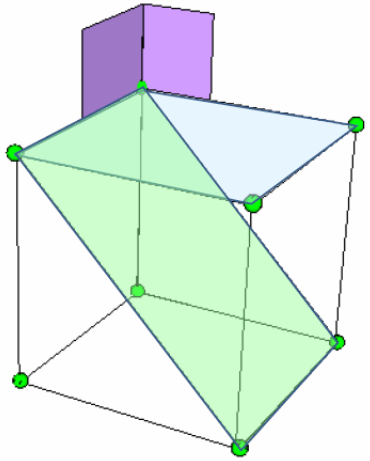
**R = a/2**

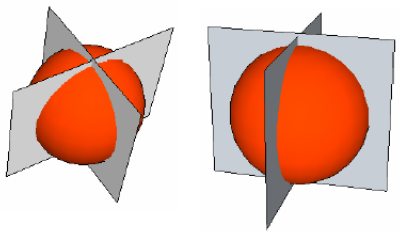
$$\mathbf{P} = \frac{1}{8} \times \left(\frac{4}{3}\right) \pi R^3 \times \frac{8\ corners}{a^3} = \frac{\pi}{6}$$

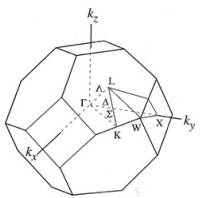


# Cubic Lattice: Surface Issues



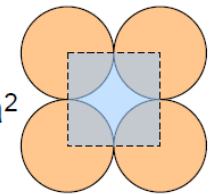





Areal Density

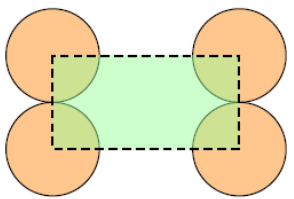
=(1/4 per corner) x (4 corners)/$a^2$

=1/$a^2$ $cm^{-2}$



Areal density (face diagonal)

= (1/4 points/corner) x (4 corners)/$\sqrt{2}a^2$ $cm^{-3}$



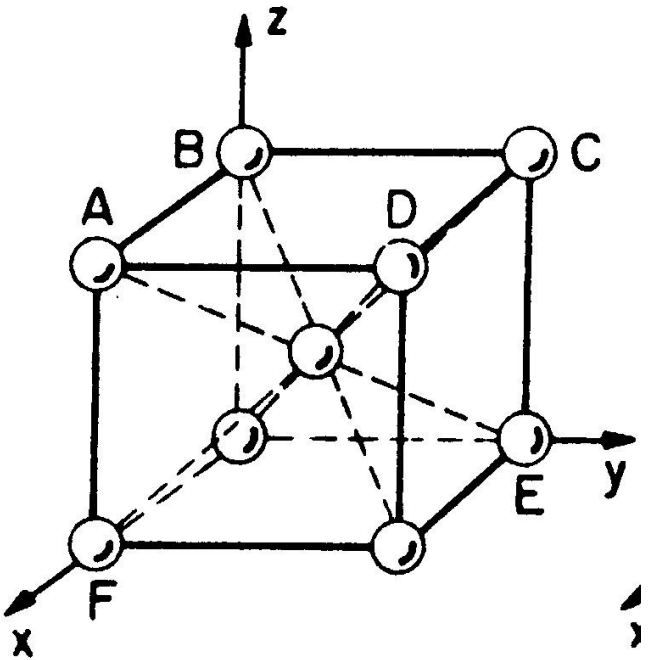
# Body Centered Cubic (BCC)

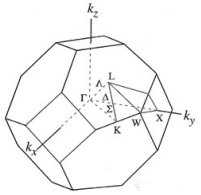
Mo, Ta, W

Coordination Number = ? 8

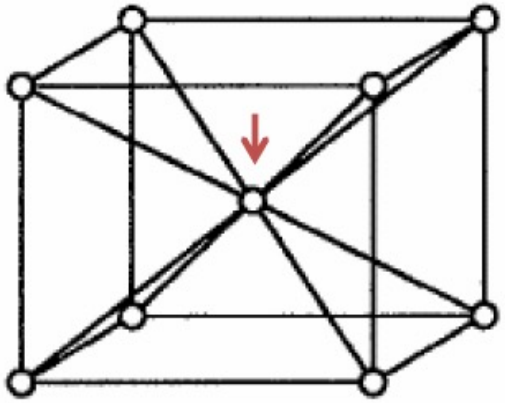
# atoms/lattice = ? 2

Packing fraction? $\frac{2 \times \frac{4}{3}\pi r^3}{a^3}$ $2r = \frac{\sqrt{3}}{2}a$

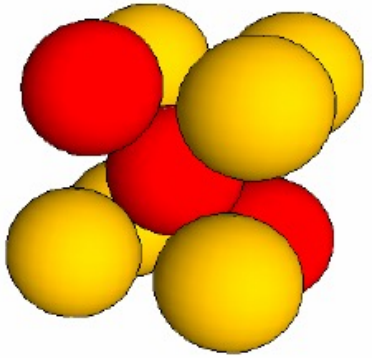


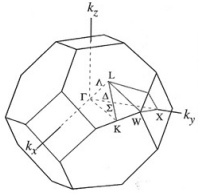
# Body Centered Cubic (BCC)



Points per cell
= 1/8 x 8 @corners
+ 1 @inside
=2

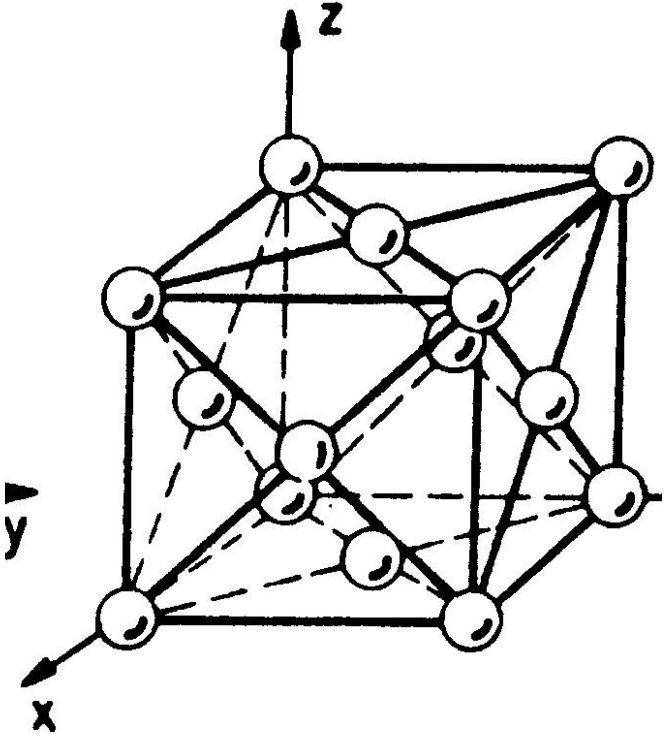


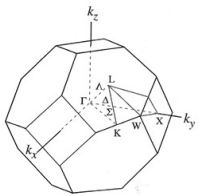
If the lattice constant a = 5Å, what is volume density?



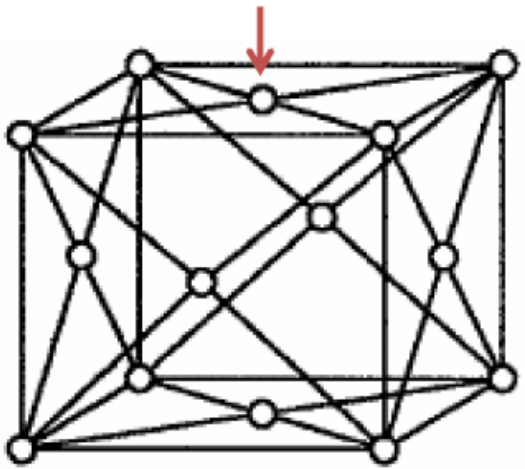
# Face Centered Cubic (FCC)

- Al, Ag, Au, Pt, Pd, Ni, Cu
- Coordination Number = ? 12
- #atoms/cell = ? 4
- Packing fraction = ? $\frac{4 \times \frac{4}{3}\pi r^3}{a^3}$ $2r \leq \frac{\sqrt{2}}{2}a$

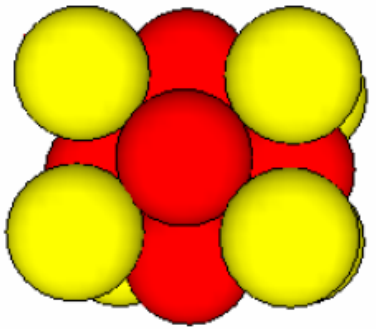


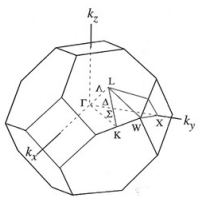



# Face Centered Cubic (FCC)



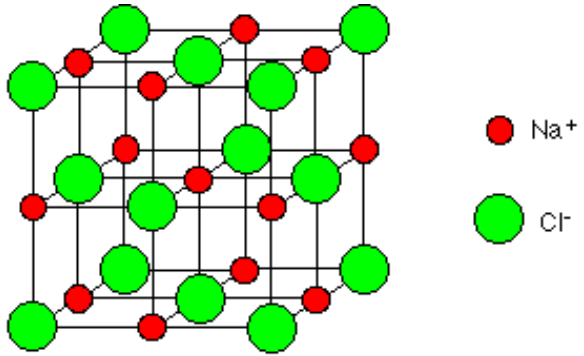
Points per cell
= 1/8 x 8 @corners
+ 1/2 x 6 @faces
=4

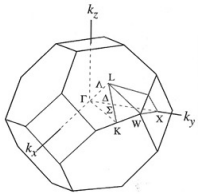




# Sodium Chloride (NaCl)

- Lattice Constant of NaCl: 5.6402Å
- Molar Mass: 58.44 g/mol
- Avogadro's number: $N_A = 6.022140857(74) \times 10^{23}\ mol^{-1}$
- Density ?

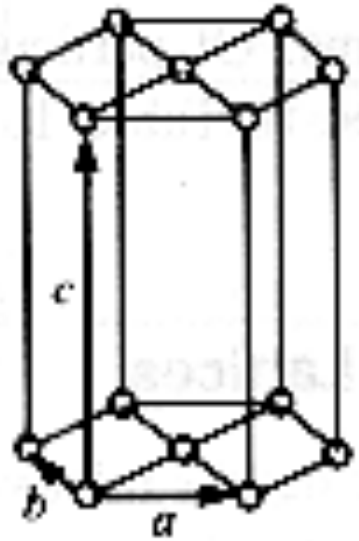


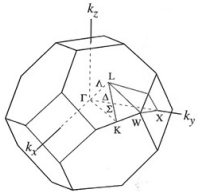
# Only other type common in ICs

**Hexagonal Lattice**

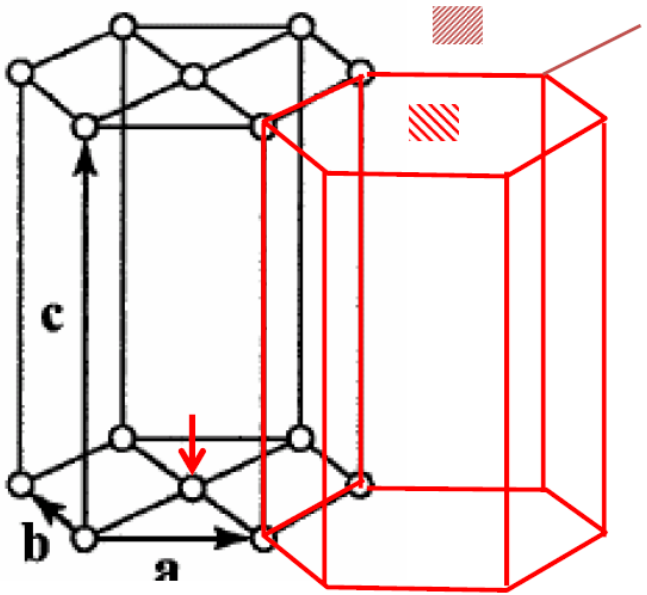
**$Al_2O_3$, Ti, other metals**



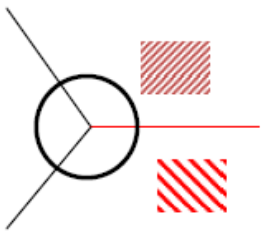

**Hexagonal**



# Hexagonal Lattice

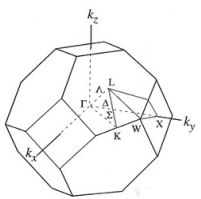



1/2 x 2 @faces =1



1/2 x 1/3 x 12 @corners =2

3 points/cell



# Diamond Structure

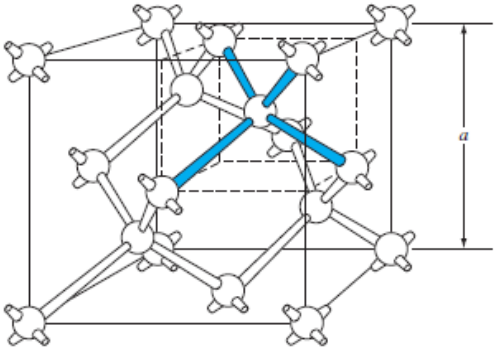



Figure 1.11 | The diamond structure.

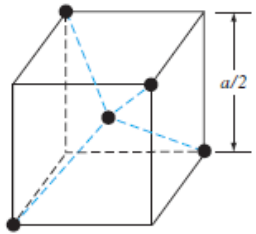



Figure 1.12 | The tetrahedral structure of closest neighbors in the diamond lattice.

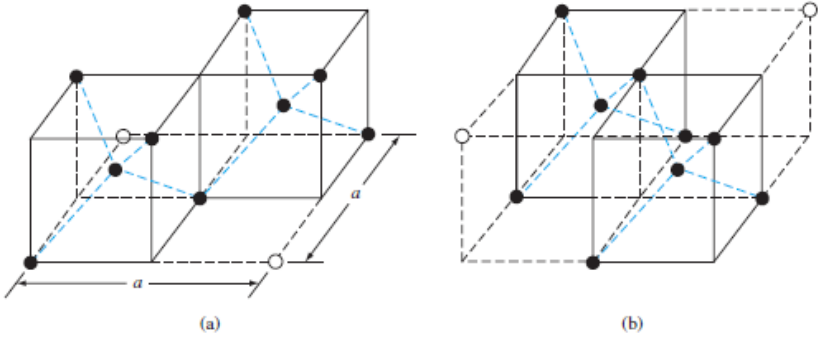
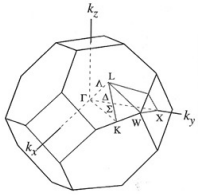
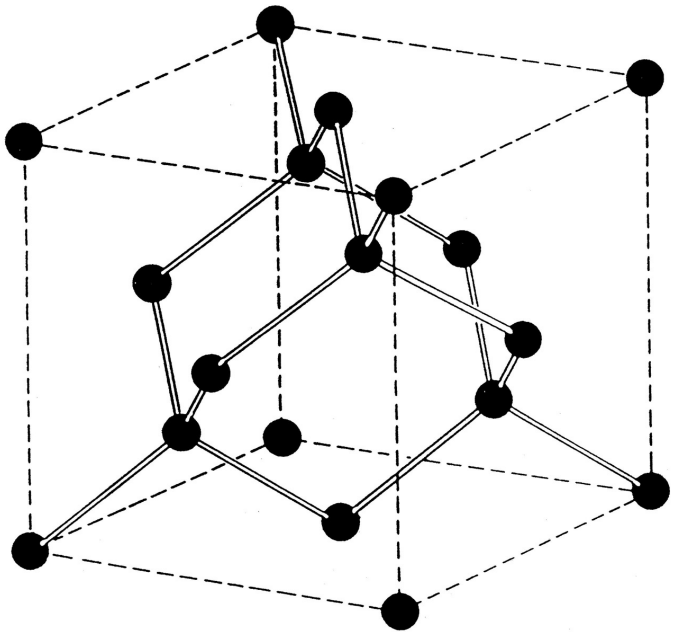


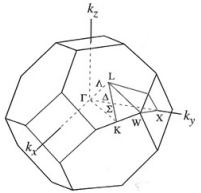

Figure 1.13 | Portions of the diamond lattice: (a) bottom half and (b) top half.



# Diamond Lattice

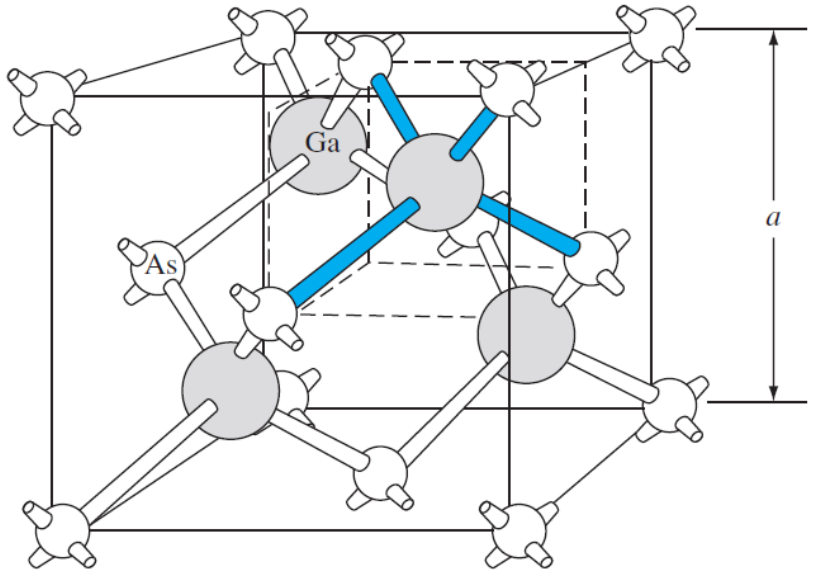
- C, Si, Ge
- a=5.43Å for Si
- Coordination Number = ? 4
- Packing fraction = ? $\frac{8\times\frac{4}{3}\pi r^3}{a^3}$ $2r \le \frac{\sqrt{3}}{4}a$
- Two FCC offset by a/4 in each direction
- FCC lattice with
- 2 atoms/site

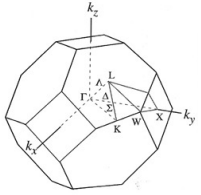




# Zinc-blende Structure

- III-V semiconductors
  - GaAs, InP,
  - InGaAs,
  - InGaAsP,……..
- For GaAs:
  - Each Ga surrounded By 4 As
  - Each As Surrounded by 4 Ga



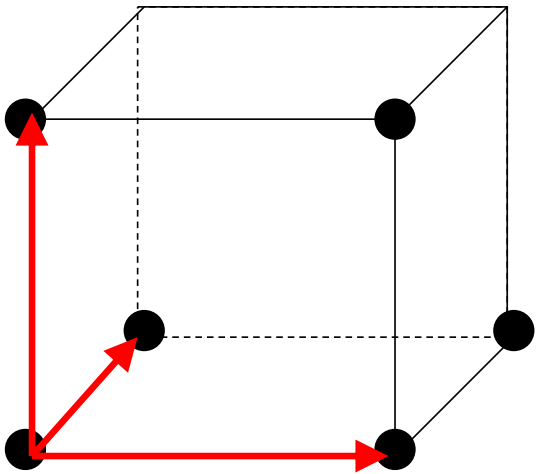
# Lattice Vectors

Simple cubic lattice

$\vec{a}$= (1,0,0)a

$\vec{b}$= (0,1,0)a

$\vec{c}$ = (0,0,1)a



bcc : $\vec{R}_1=[-\frac{1}{2},\frac{1}{2},\frac{1}{2}]a$

$\vec{R}_2=[\frac{1}{2},-\frac{1}{2},\frac{1}{2}]a$

$\vec{R}_3=[\frac{1}{2},\frac{1}{2},-\frac{1}{2}]a$

fcc : $\vec{R}_1=[0,\frac{1}{2},\frac{1}{2}]a$

$\vec{R}_2=[\frac{1}{2},0,\frac{1}{2}]a$

$\vec{R}_3=[\frac{1}{2},\frac{1}{2},0]a$

+ (0,0,0)

$(\frac{1}{4},\frac{1}{4},\frac{1}{4})a$

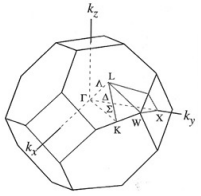
⇓

Diamond Lattice

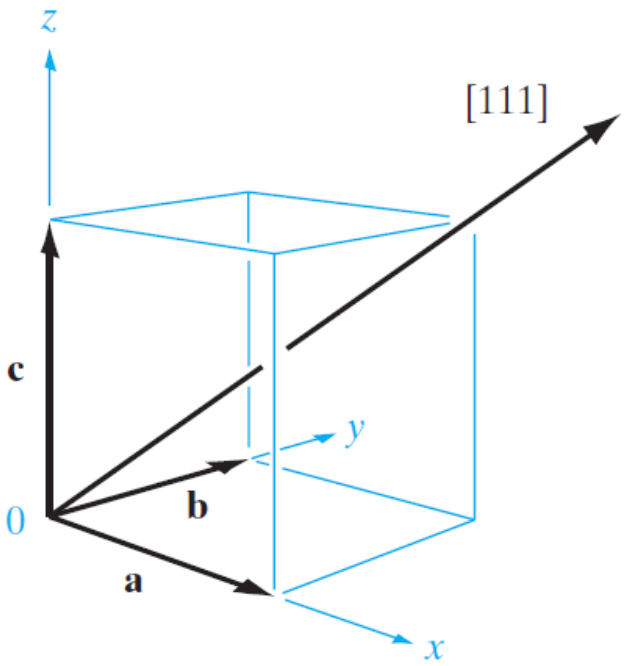
Three primitive vectors are 'coordinates' in terms of which all lattice coordinates R can be expressed

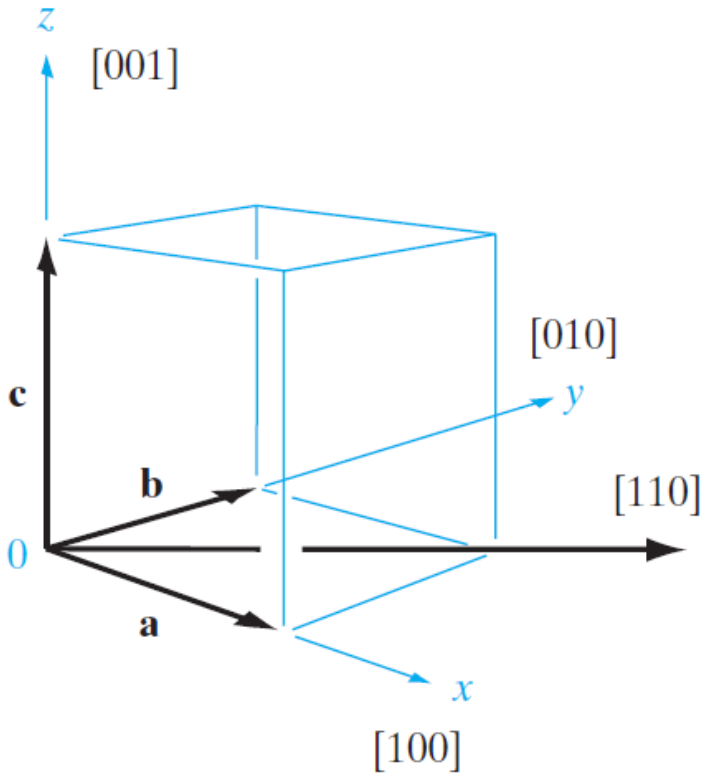
$\vec{R}$ = m$\vec{a}$ + n$\vec{b}$+ p$\vec{c}$ (m,n,p: integers):[mnp]

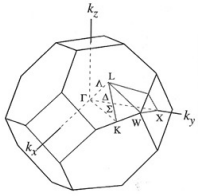
Equivalent direction indices <mnp>



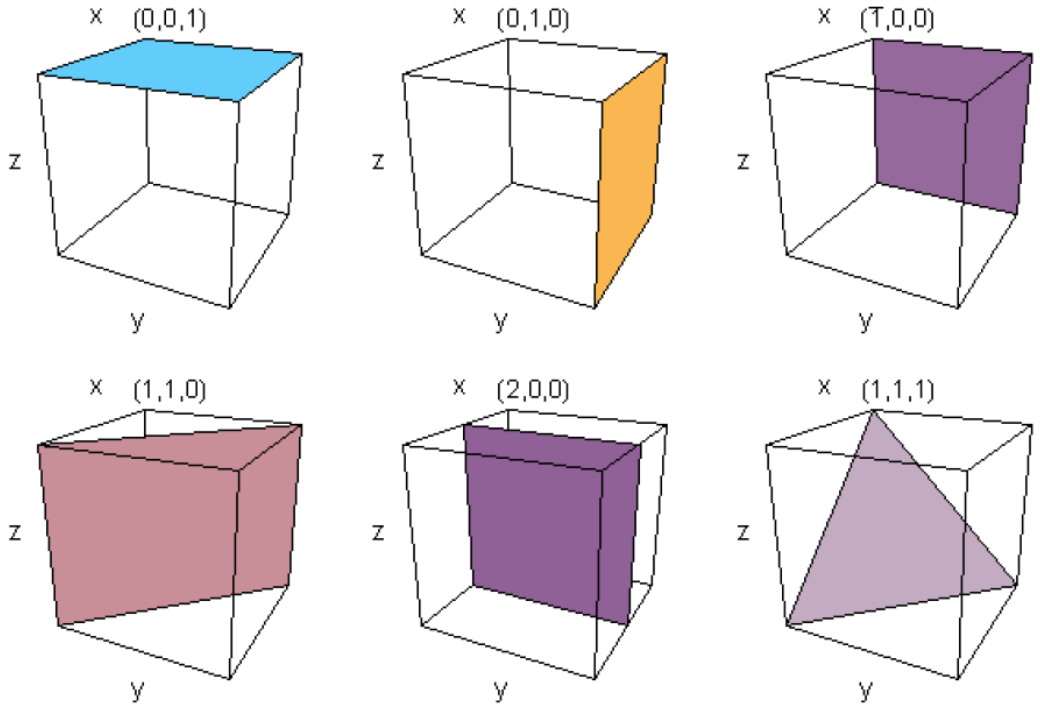
# Crystal Direction

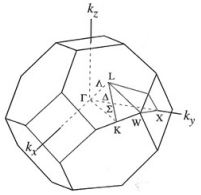


# Planes in a Crystal

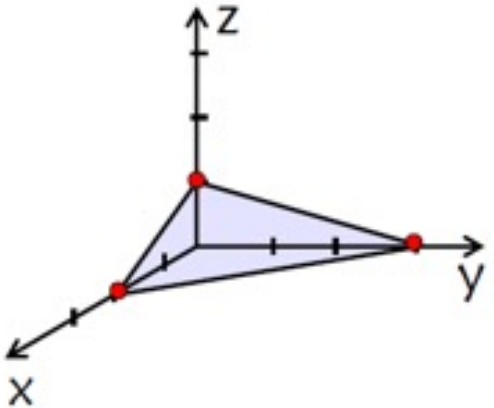


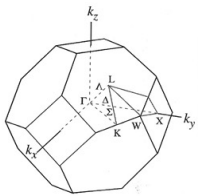



**Crystal Planes denoted by *Miller Indices* h,k,l**

# Miller Indices of a Plane

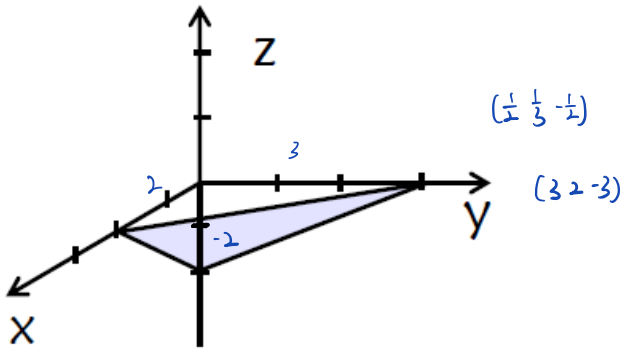
- Set up axes along the edges of unit cell
- Normalize the intercepts… 2, 3, 1
- Invert/rationalize intercepts….1/2, 1/3, 1
  - ➢3/6, 2/6, 6/6
- Enclose the numbers in curvilinear brackets
  - ➢(326)



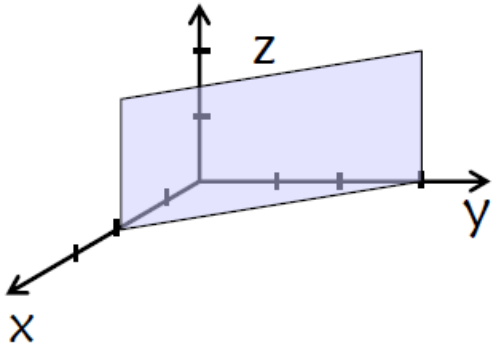
# Few More Rules…

## Negative Intercept

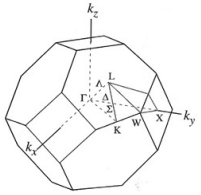



| | | |
|---|---|---|
| 2, | 3, | -2 |
| 1/2, | 1/3, | -1/2 |
| 3, | 2, | -3 |
| $(32\bar{3})$ | | |

## Intercept at infinity

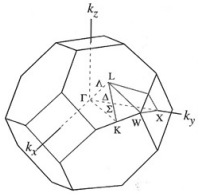



| | | |
|---|---|---|
| 2, | 3, | $\infty$ |
| 1/2, | 1/3, | 0 |
| 3, | 2, | 0 |
| (320) | | |

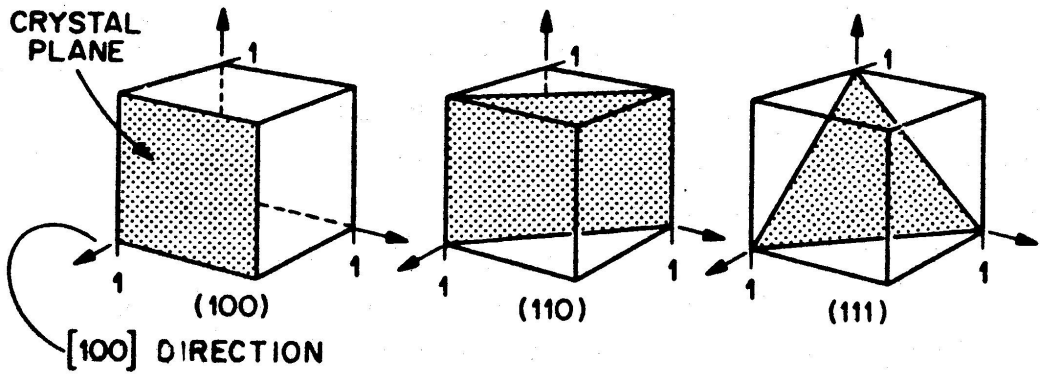


# Crystallographic Notation

| Notation | Interpretation |
|---|---|
| $(h\,k\,l)$ | crystal plane |
| $\{h\,k\,l\}$ | equivalent planes |
| $[h\,k\,l]$ | crystal direction |
| $<h\,k\,l>$ | equivalent directions |



# Some Prominent Planes

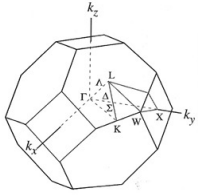



**Equivalent planes denoted by {}**

**{100}=(100), (010), (001)**

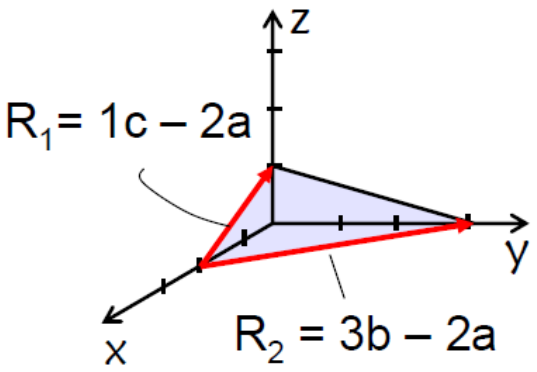
**For Cubic structures:**

**[h,k,l] ⊥ (h,k,l)**



# Where does Miller Indices Come From?

Miller indices: (326)



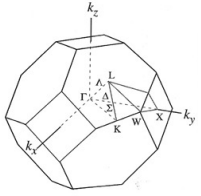

**For cubic crystals, a plane and the direction normal to plane have the same indices**

Normal to the surface

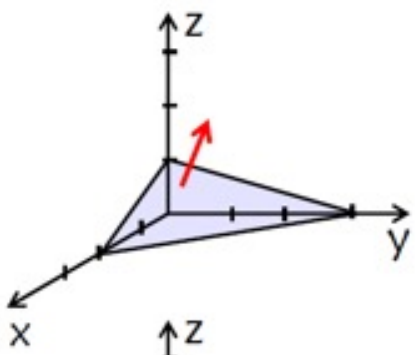
$$R_2 \times R_1 = \begin{vmatrix} a & b & c \\ -2 & 3 & 0 \\ -2 & 0 & 1 \end{vmatrix} = 3a + 2b + 6c$$

Vector indices same as Miller indices !

(326) vs. [326]



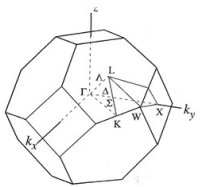
# Two Useful Relationship




The distance d between two adjacent planes labeled (hkl) is given in term of the lattice constant, a, as $d = a/(h^2 + k^2 + l^2)^{1/2}$, useful for XRD

The angle $\theta$ between two different Miller index directions is given by

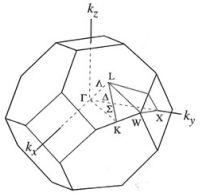
$$cos\theta = \{h_1h_2 + k_1k_2 + l_1l_2\}/\{(h_1^2 + k_1^2 + l_1^2)^{1/2}(h_2^2 + k_2^2 + l_2^2)^{1/2}\}$$



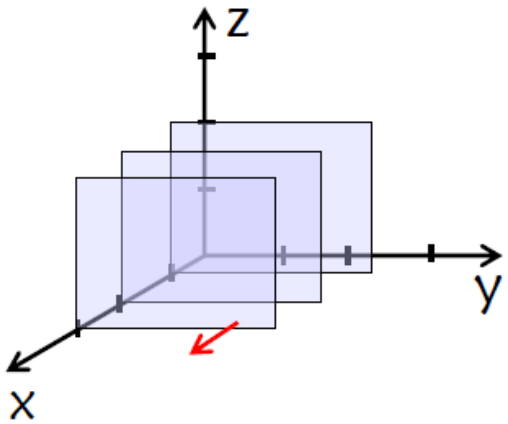
# Why Bother Naming Planes?

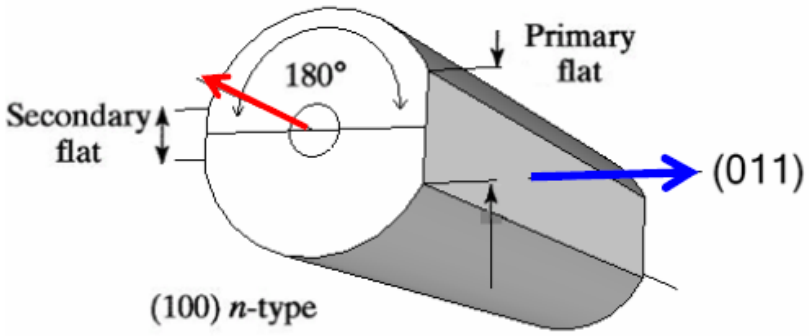
- Fabrication motivations
  - Certain planes cleave easier
  - Wafers grown and notched on specific planes
  - Pattern alignment

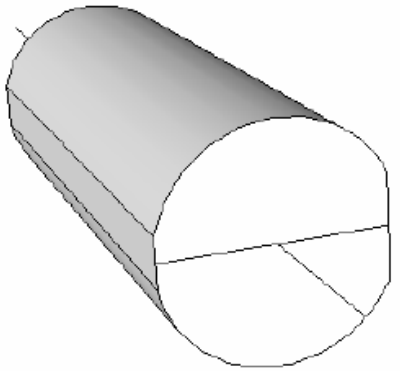
- Chemical/Material Motivations
  - Density of electrons different on planes
  - Reconstruction causes different environments
  - Defect densities, chemical bonding depend on orientation

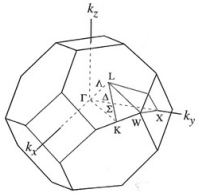


# Crystallographic Planes and Si Wafers



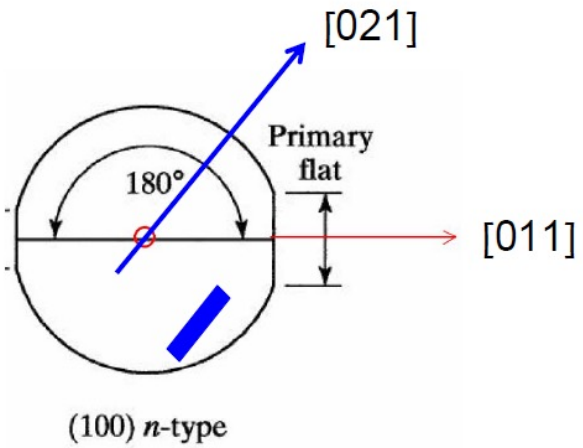



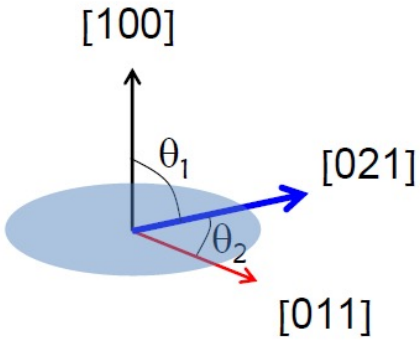



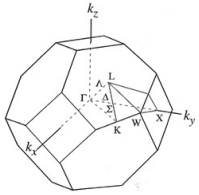


# Example Use of Miller Indices

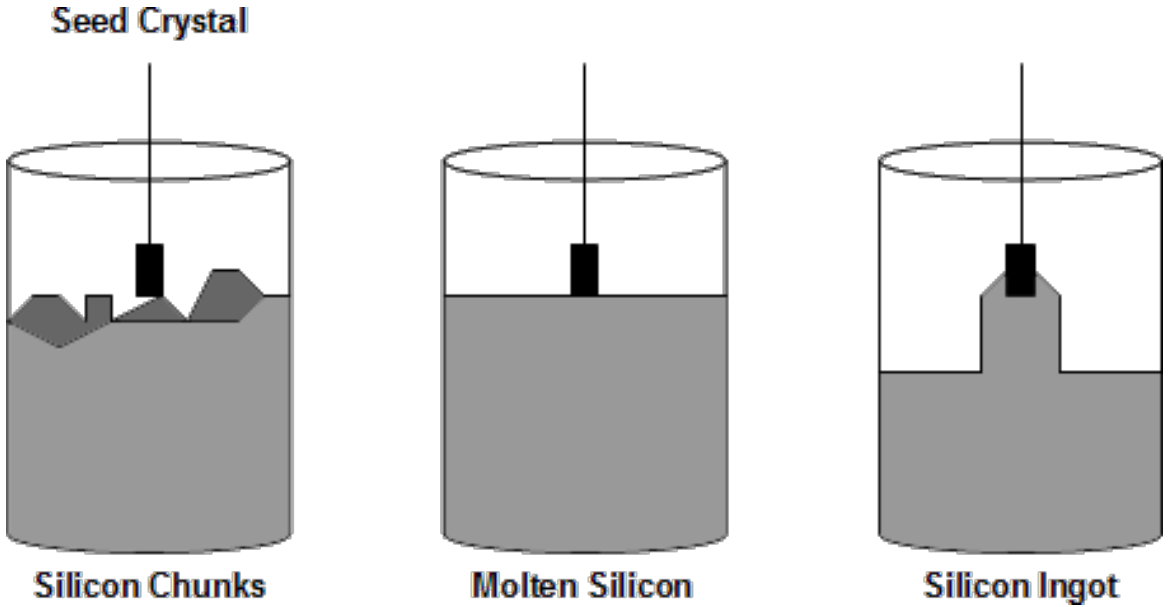
- Find [021] Direction

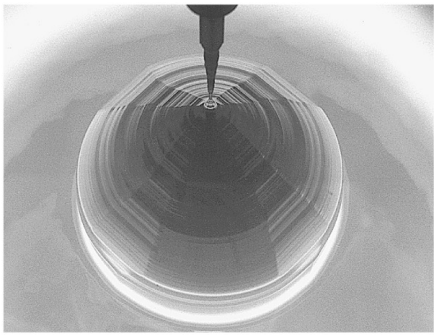


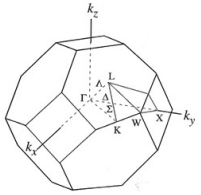
# Where do (Pure) Si Wafers Come From?





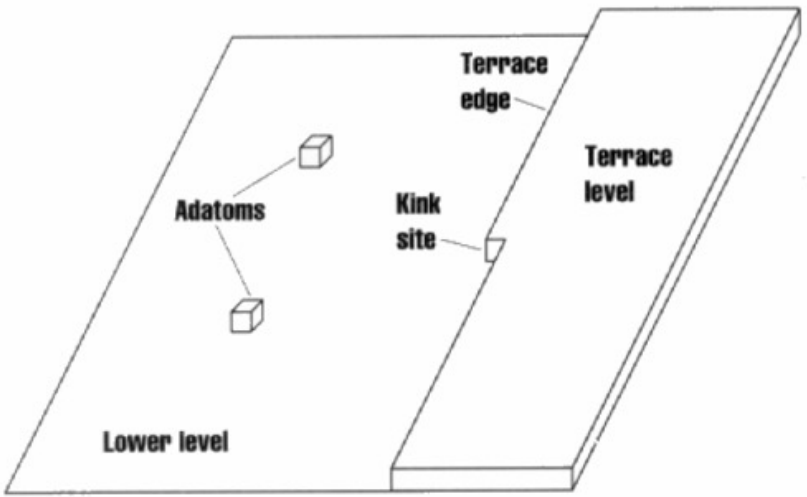
*Wikipedia*

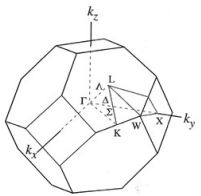


(99.9999999% purity)

# How About GaAs and Other III-V?

- Epitaxial Growth
  - MOCVD: Metal-organic Chemical Vapor Deposition
  - MBE: Molecular beam epitaxy



# Epitaxy

- Importance of lattice mismatch
- The lattice constant of the epitaxially grown layer needs to be close to the lattice constant of the substrate wafer. Otherwise the bonds can not stretch far enough and dislocations will result.

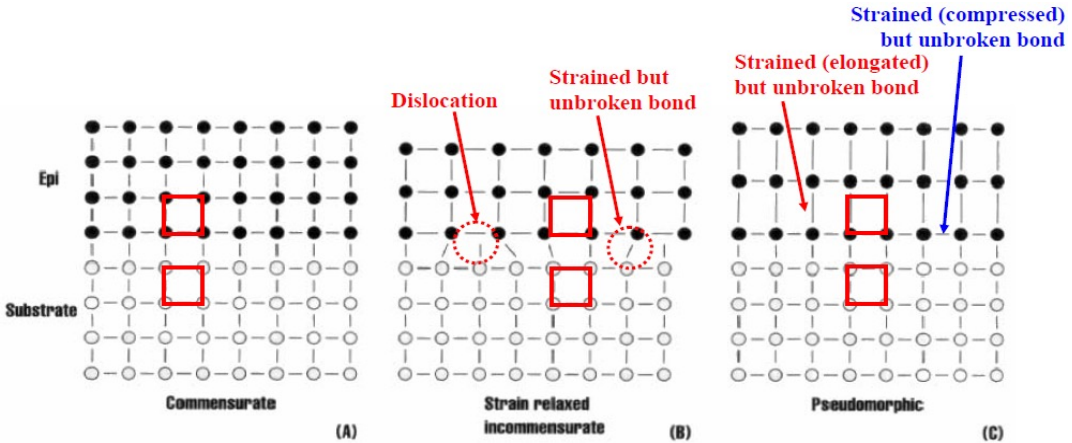
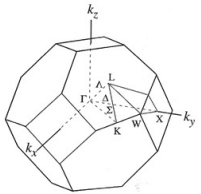
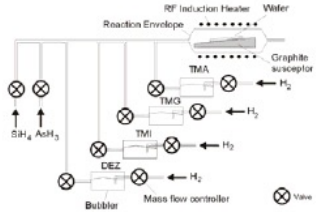



**Figure 14-15** Epitaxial growth processes can be divided into (a) commensurate, (b) strain relaxed incommensurate, and (c) incommensurate but pseudomorphic.

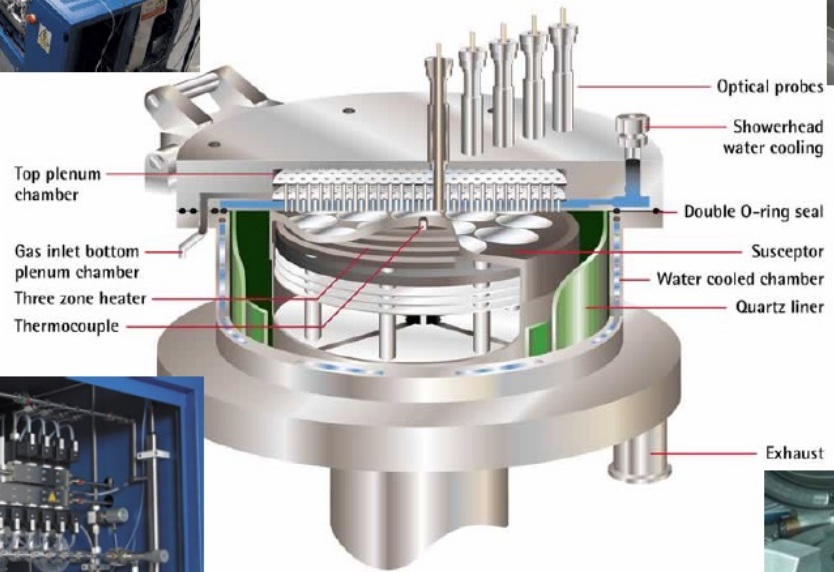


# Commercial Thomas Swan® MOCVD



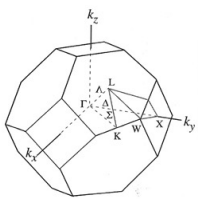








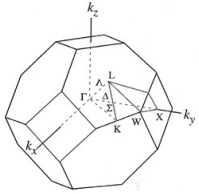



# MBE

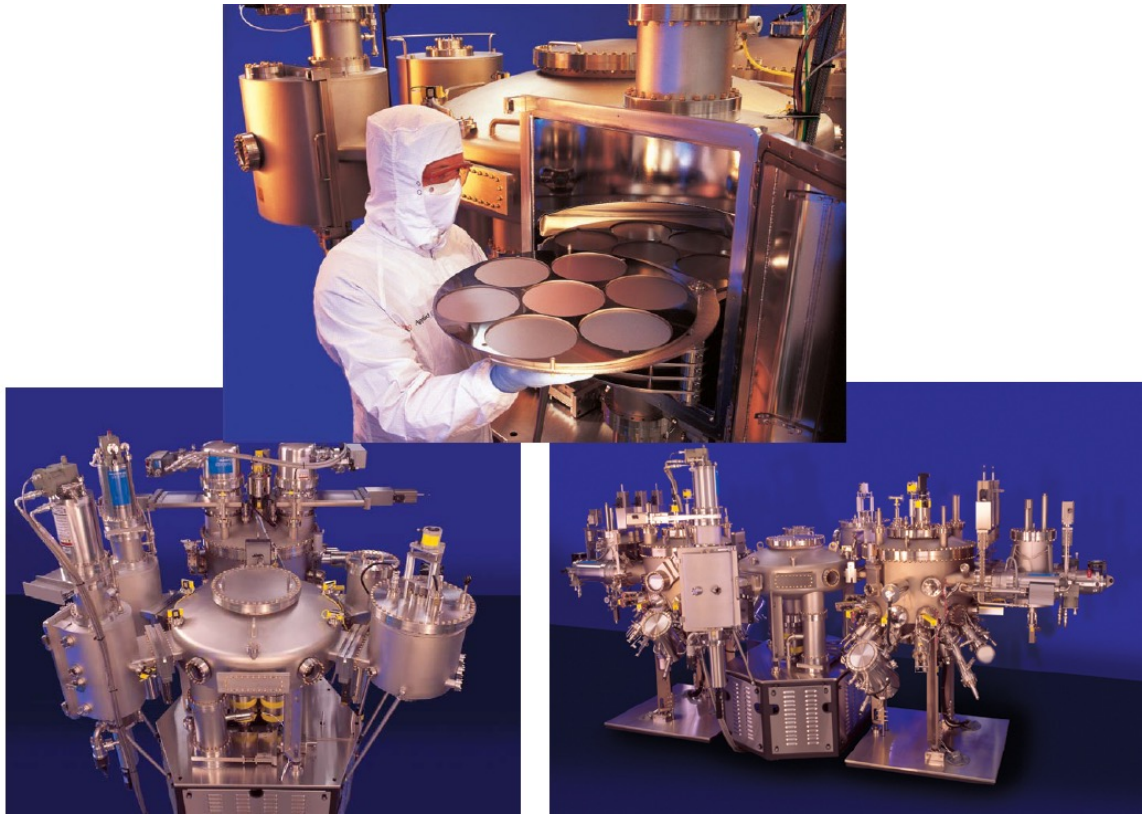
- Offers the highest purity material (due to UHV conditions) and the best layer control (almost any fraction of an atomic layer can be deposited and layers can be sequenced one layer at a time (for example Ga then As then Ga etc…).

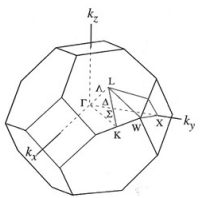
卓以和（**Alfred Y. Cho**）





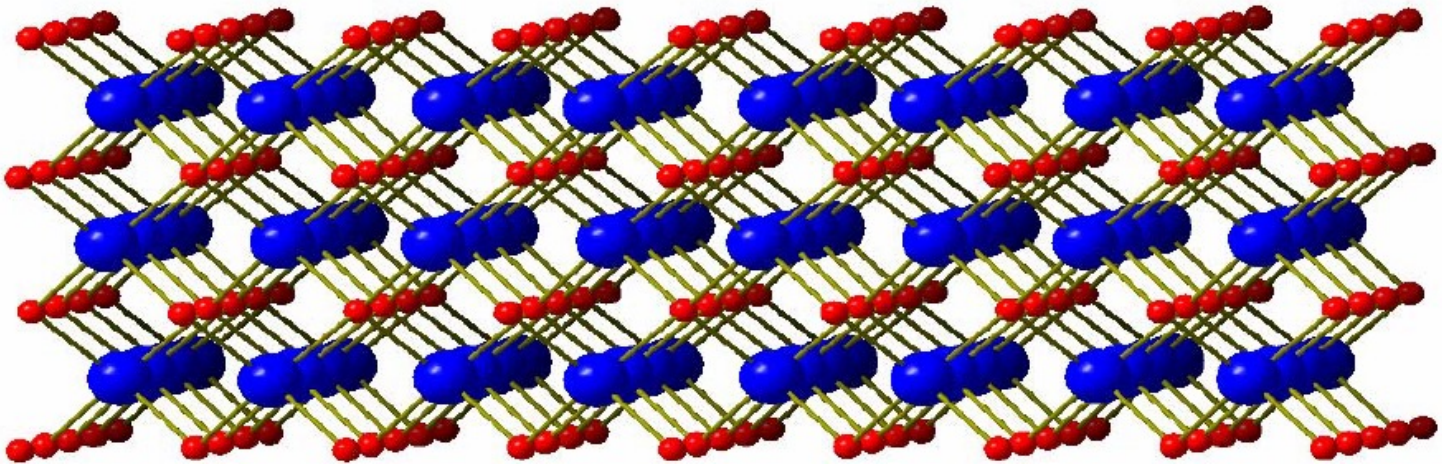
# MBE

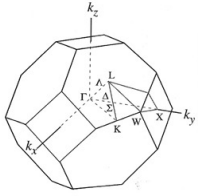




# Epitaxy

- Repeating a crystalline structure by the atom by atom addition.
- Chemistry controls the epitaxy to insure that, for example, Ga bonds only to N and not Ga-Ga or N-N bonds





# Heterostructure

